

# Factory Operations Report 2025

## 1. Executive Summary

This report reviews the 2025 operating performance of the Northfield assembly plant. Overall production efficiency, measured as overall equipment effectiveness (OEE), peaked at 92% in Q2 and ended the year at 81% in Q4. Total output for the year was 506,000 units. Management remains committed to the annual target of 85% average efficiency.

The plant operates three production lines across two shifts. Line 3, which produces the high-volume gearbox housing, accounts for roughly 45% of total output and therefore dominates the plant-level results. Safety performance was stable with no lost-time incidents recorded in the year.

## 2. Plant Overview

Northfield employs 240 people, of whom 168 work directly on the production lines. Investment in 2025 focused on automation of the packaging area and on a new quality inspection station at the end of Line 2. The product mix was unchanged from 2024, so differences between quarters are attributable to operating conditions rather than mix.

### 3. Quarterly Production Efficiency

Table 1 gives the quarterly efficiency, output and scrap rate. Efficiency is the product of availability, performance and quality as defined in the plant KPI handbook.

| Quarter | Efficiency (%) | Output (units) | Scrap rate (%) |
|---------|----------------|----------------|----------------|
| Q1      | 78             | 118,000        | 3.1            |
| Q2      | 92             | 139,000        | 2.2            |
| Q3      | 85             | 128,000        | 2.6            |
| Q4      | 81             | 121,000        | 3.0            |

Table 1: Quarterly production efficiency, output and scrap rate, 2025.

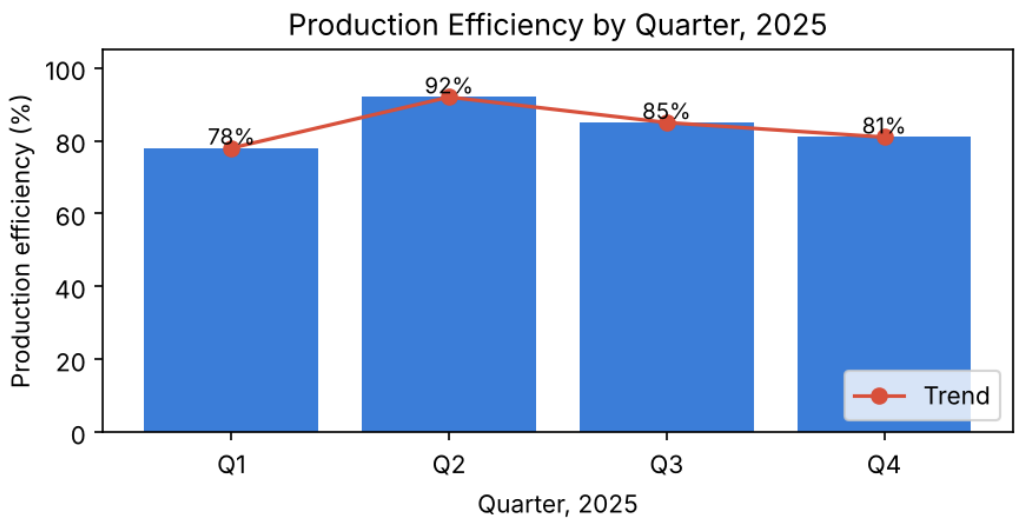


Figure 1: Production efficiency by quarter (%), bar chart with trend line.

The chart shows efficiency rising from 78% in Q1 to a peak of 92% in Q2, followed by a steady decline through Q3 (85%) and Q4 (81%).

## 4. Analysis of the Q2 to Q4 Decline

Efficiency fell by 11 percentage points between Q2 (92%) and Q4 (81%), a relative decline of about 12.0%. Three main reasons explain the change:

**Reason 1 - Rising mechanical failures on Line 3.** Ageing spindle bearings caused repeated unplanned stops. Mechanical failure downtime grew from 28 hours in Q2 to 71 hours in Q4 (see the Maintenance Summary 2025).

**Reason 2 - Raw material shortages.** A supplier delay for cast aluminium blanks starved the lines in the second half. Material shortage downtime rose from 9 hours in Q2 to 38 hours in Q4.

**Reason 3 - Staffing turnover and training.** Eleven experienced operators left in Q3 and were replaced by trainees, which lowered line speed and raised the scrap rate from 2.2% in Q2 to 3.0% in Q4.

## 5. Outlook and Recommendations

For 2026 the plant will replace the Line 3 spindle bearings in the Q1 planned shutdown, dual-source the aluminium blanks, and introduce a structured four-week onboarding program. These measures target a return to at least 88% efficiency by Q2 2026.